

SHREYA BHAT

GenAI Engineer | LLMs , Agentic AI , Prompt Engineering

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SUMMARY

GenAI Engineer with hands-on experience building LLM-powered pipelines, agentic workflows, and multimodal document-processing systems. Skilled in prompt engineering, RAG, and agentic frameworks (LangChain, LangGraph, CrewAI, ReAct) to build and evaluate production-oriented AI solutions. Proficient in Python, with working knowledge of ML/DL fundamentals.

EXPERIENCE

Data Scientist | CIPLA Ltd

July 2024 - June 2025

- Built a RAG-based document processing system (LangChain + pgvector) indexing 1,000+ equipment logs and SOPs, reducing monthly troubleshooting job cards from 120+ to ~40 (~67% reduction).
- Applied prompt engineering to tune retrieval and response quality across multiple LLM-backed workflows.
- Evaluated RAG pipeline outputs periodically against a 100-observation golden dataset using RAGAS metrics (faithfulness, context recall, answer relevancy) to evaluate and improve model performance, accuracy, and reliability.
- Built an anomaly detection pipeline across 1,400+ sensors (15 SCADA, 47 BMS systems), integrated with a knowledge retrieval system to cut root-cause analysis time by 55–60%.

PROJECTS

Multimodal Financial Document Understanding System | *Multimodal LLMs (LLaVA, Qwen-VL), LangGraph, CrewAI, ReAct*

- Built a multimodal document-understanding pipeline using open-source vision-language models (LLaVA, Qwen-VL) to extract text, tables, charts, signatures, and layout from multi-page financial documents (e.g., bank statements) — replacing traditional OCR-based extraction.
- Orchestrated the multi-stage extraction, validation, output pipeline using LangGraph.
- Built a CrewAI-based validation agent using the ReAct pattern to verify extracted fields (transactions, totals, anomalies) before finalizing output.
- Structured outputs into machine-readable JSON (account holder, account number, transactions, credits/debits, anomalies) for downstream processing.

AI Hallucination Detection System | *LLM Evaluation, RAG, FAISS, FastAPI, Multi-stage*

- Built an LLM validation pipeline performing claim extraction, web evidence retrieval, and semantic similarity scoring to detect hallucinations.
- Integrated Google Gemini API and FAISS vector search for real-time evidence grounding, achieving ~87% detection accuracy.
- Designed an LLM observability layer providing trust scoring and reliability monitoring for AI-generated outputs, deployed via FastAPI with an interactive Streamlit dashboard.

AskQL - Natural Language to SQL Query System | *NLP, SQL, LLM, FastAPI, React*

- Built an AI-powered natural language-to-SQL system converting plain English queries into structured SQL commands, using rule-based and NLP-driven pipelines.
- Applied classical NLP techniques (tokenization, entity/intent extraction) for query understanding, with exploratory LLM-based enhancements and a modular architecture for future generative AI integration.

EDUCATION

B.E. - Electronics and Instrumentation Engineering

Nov 2020- May 2024

Bangalore institute of technology, Bengaluru

PGP in Data Science with specialization in Gen AI

June 2025 - May 2026

Great learning

SKILLS

Generative AI: LLMs, Prompt Engineering / Prompt Tuning, RAG, Multimodal LLMs, Small Language Models (SLMs), Hugging Face Transformers, FAISS, Vector Databases, LLM Fine-tuning

Agentic Frameworks: LangChain, LangGraph, CrewAI, ReAct

LLM Evaluation & Monitoring: RAGAS (faithfulness, context recall, answer relevancy), Semantic Similarity Scoring

Languages: Python, SQL

ML & AI: Scikit-learn, XGBoost, PyTorch (basic), TensorFlow, Deep Learning

Data Processing: Pandas, NumPy, Feature Engineering, Statistical Analysis

Backend & Deployment: FastAPI, Flask, Streamlit, Docker, REST APIs

Databases & Tools: MySQL, SQLite, PostgreSQL, FAISS, Git, GitHub